systems humans had already es tabished on the planet it would also constitute the irresponsible destruction of a unique scientific resource and database, the Martian surface.

What about other greenhouse gases? Alternatively, we might take chlorofluorocarbons (CFCs or HCFCs) to Mars after manufacturing them on Earth. These are artificial substances that, so far as we know, are found nowhere else in the Solar System. We can certainly imagine manufacturing enough CFCs on Earth to warm Mars, because by *accident* in a few decades with present technology on Earth we've managed to synthesize enough to contribute to global warming on our planet. Transportation to Mars would be expensive, though: Even using *Saturn V*- or *Energiya*-class boosters, it would require at least a launch a day for a century. But perhaps they could be manufactured from fluorine-containing minerals on Mars.

There is, in addition, a serious drawback: On Mars as on Earth, abundant CFCs would prevent formation of an ozone layer. CFCs might bring Martian temperatures into a clement range, but guarantee that the solar ultraviolet hazard would remain extremely serious. Perhaps the solar ultraviolet light could be absorbed by an atmospheric layer of pulverized asteroidal or surface debris injected in carefully titrated amounts above the CFCs. But now we're in the troubling circumstance of having to deal with propagating side effects, each of which requires its own large-scale technological solution.

A third possible greenhouse gas for warming Mars is ammonia (NH_3). Only a little ammonia would be enough to warm the Martian surface to above the freezing point of water. In principle, this might be done by specially engineered microorganisms that would convert Martian atmospheric N_2 to NH_3 as some microbes do on Earth, but do it under Martian conditions. Or the same conversion might be done in special factories. Alternatively, the nitrogen required could be carried to Mars from elsewhere in the Solar System. (N_2 is the principal constituent in the atmospheres of both Earth and Titan.) Ultraviolet light would convert ammonia back into N_2 in about 30 years, so there would have to be a continuous resupply of NH_3.